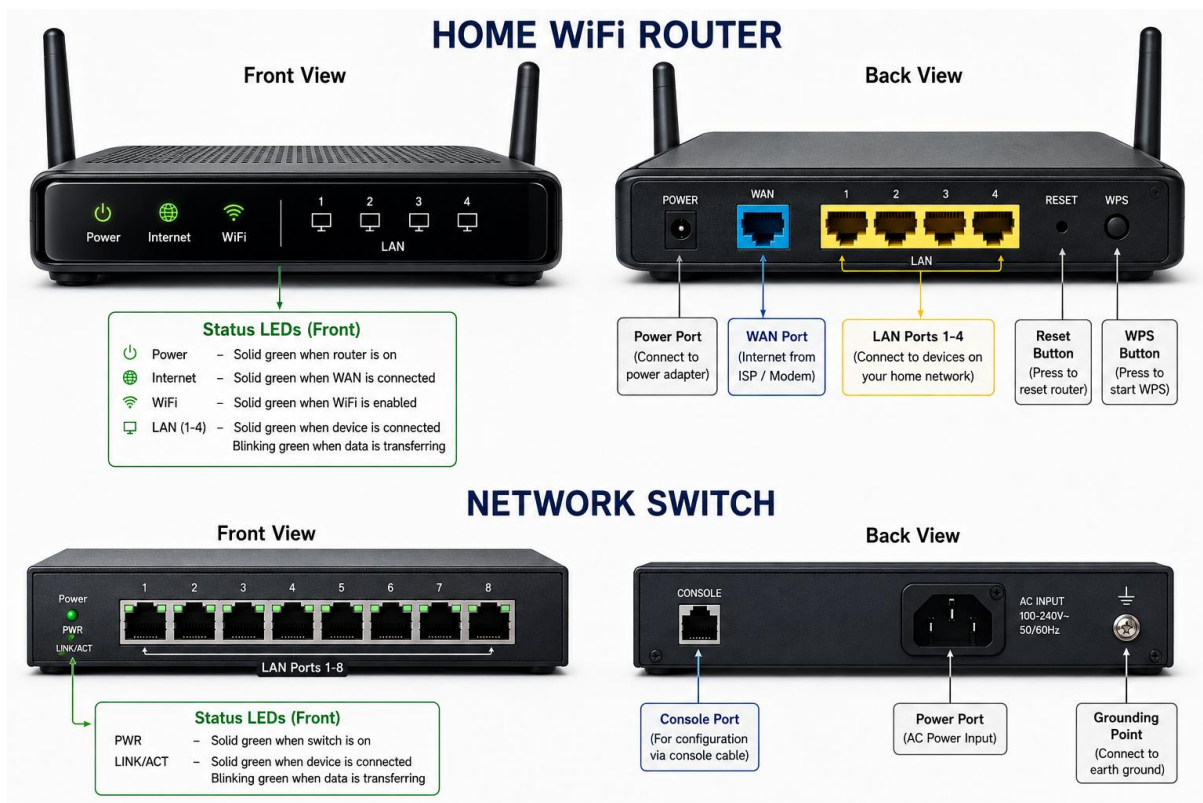


Assignment – 3.2

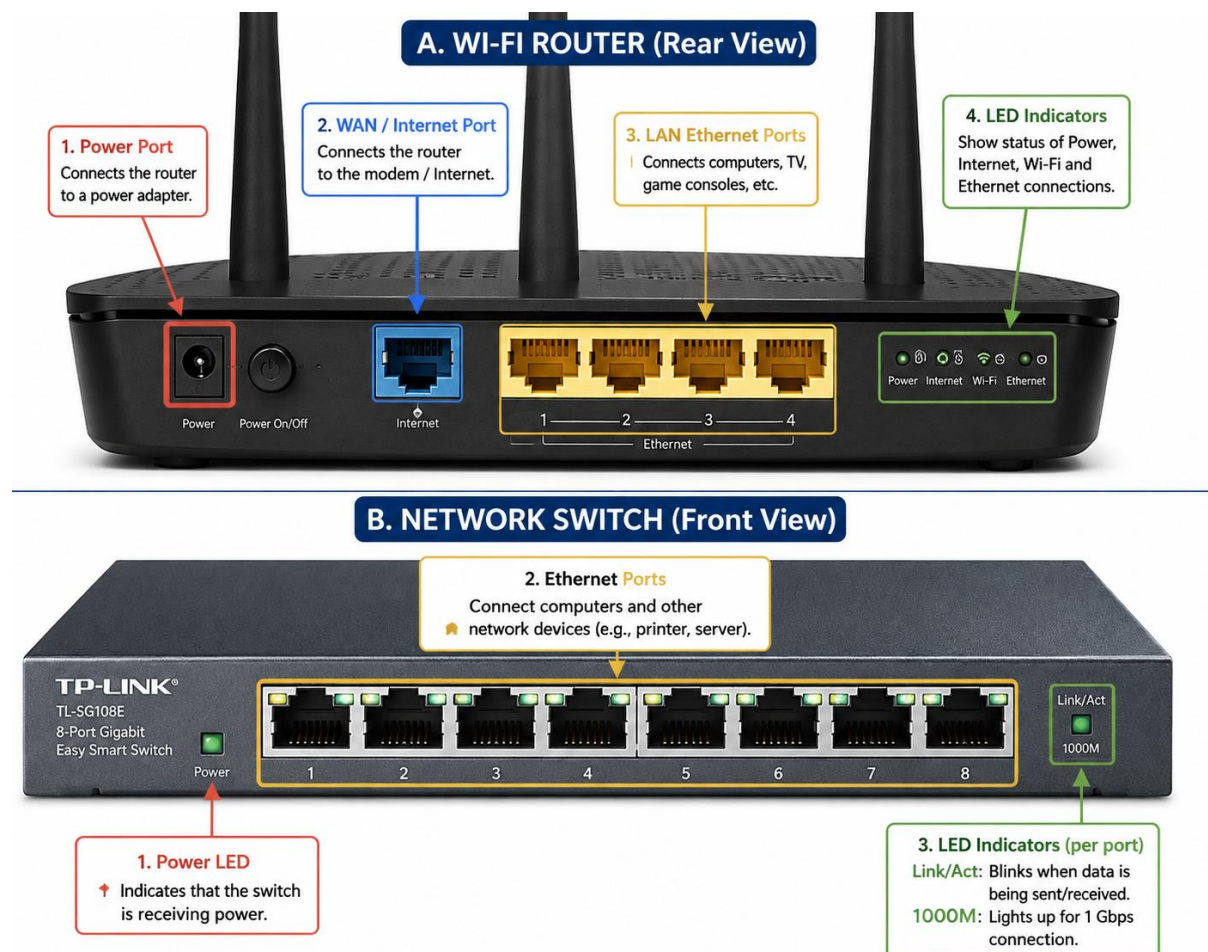
Q - 1. Draw a labelled diagram of a typical home Wi-Fi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



Q - 2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Feature	Router	Switch
1. Main Function	Connects different networks together, such as a home network to the Internet.	Connects devices within the same network, such as computers and printers.
2. Address Used	Mainly uses IP addresses to decide where to send packets.	Mainly uses MAC addresses to forward Ethernet frames.
3. Traffic Handling	Routes traffic between different networks and chooses the appropriate path.	Sends traffic to the specific device/port where the destination device is connected.

Q - 3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



A. Wi-Fi Router

- **Power Port:** Used to connect the router to an electrical power adapter.
- **WAN/Internet Port:** Connects the router to the Internet/modem.
- **LAN Ethernet Ports:** Used to connect computers, TVs, or other devices using Ethernet cables.
- **LED Indicators:** Show Power, Internet, Wi-Fi, and LAN activity/status.

B. Network Switch

- **Power Port:** Supplies electrical power to the switch.
- **Ethernet Ports:** Used to connect multiple computers and network devices together.
- **LED Indicators:** Usually show Power, Link, and Data/Activity for each port.

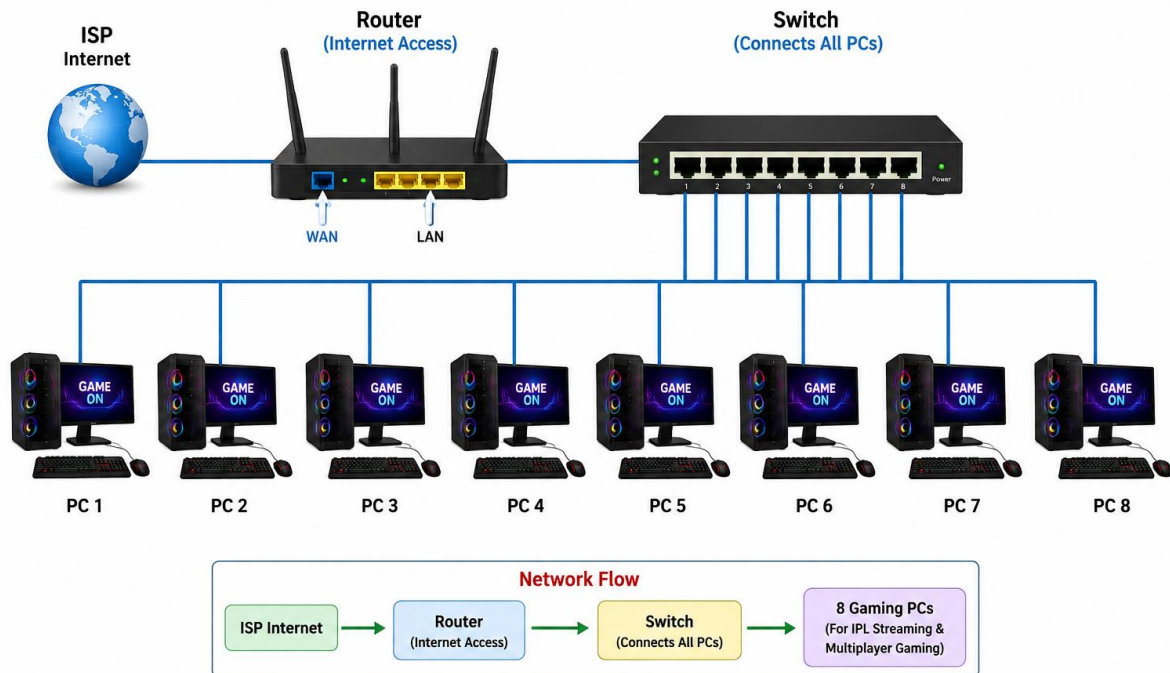
Q - 4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Gaming Café Network Scenario

Imagine I am setting up a small gaming café with 8 gaming PCs for IPL streaming and multiplayer games.

- First, I would connect the ISP Internet line to the router.
- The router connects the café's network to the Internet and manages IP addresses.
- Then, I would connect the router to a network switch using an Ethernet cable.
- Finally, I would connect all 8 gaming PCs to the switch using Ethernet cables.

Gaming Café Network Setup



Justification:

- **Router:** Connects the café to the Internet and manages IP addresses.
- **Switch:** Connects all 8 PCs to the same local network.
- **Ethernet cables:** Provide a fast and stable connection for gaming and IPL streaming.